Activity 2.2 – Vending Machine Controller

1.Design Approach:

Designed a vending machine controller using Verilog HDL. The system accepts coin inputs, calculates the inserted amount and generates a dispense signal when the required amount is reached. Change is calculated when the inserted amount exceeds the product price.

2.Verification Steps:

- Created a separate testbench.
- Applied reset to initialize the controller.
- Tested different coin combinations.
- Verified amount accumulation.
- Checked dispense signal generation.
- Tested change calculation.
- Simulated using ModelSim.

3.Results:

The testbench checks coin accumulation, dispensing and change generation. PASS/FAIL messages are displayed.

4.Issues Faced:

Coin values must be accumulated correctly. Dispensing must occur only when the required amount is reached. Change calculation and reset conditions require proper handling.

5.Conclusion:

Successfully designed and verified a vending machine controller using Verilog HDL.